

****All exercises & prehab series mentioned here can be found on the SSDL training tab**

****If you experience pain with any of the movements, we recommend you meet with a sports medicine professional**

Test explanations & results:

- **1. Overhead Raise - Against wall** - This will measure how effective the athlete is at creating overhead motion. With this age range, we would love to see athletes have a fluid overhead position so they can move as efficiently in & out of the water as possible.
 - This test will be scored on the following criteria:
 - Good - your hands were able to touch the wall without your low back excessively flaring
 - Average - your hands were able to break the plane of your forehead without your low back excessively flaring
 - Below average - your hands were unable to break the plane of your forehead
- **What to do based on your results**
 - **If you scored a Good:**
 - We'd recommend being consistent with warming up the muscles of your shoulders before training to make sure you can maintain that range of motion.
 - For this age range, exercises we like to warmup the shoulders are:
 - Bear crawl variations, downward dogs, inch worms, band pullaparts, t-spine open books, t-spine windmills
 - **If you scored an Average or below average:**
 - There is some room for improvement with your shoulders! On top of the shoulder exercises you do as part of the dryland already, it'd be a very good idea to spend something like 5-10 minutes each day (before practice or at home) doing some exercises that build resilient & mobile shoulders. We want the swimmers in this age range to build good habits now, so as they get older and start to swim more yards they can swim as efficiently as possible.
 - Exercises
 - Quadruped reach & rotate, t-spine open books, t-spine windmills, serratus crawls, glute bridge w/ reach, halo variations

- **2. Overhead Squat** - This assessment looks at how the entire body moves as one unit. Think of the body as one big chain, with each joint (shoulders, hips, ankles) being its own respective link in the chain.
 - **What to keep an eye out for:**
 - Did the athlete maintain an overhead position through the entire motion, or did their arms come forward as they descended?
 - Did the athlete's mid back round as they descended?
 - Did the athlete's knees stay in line with their toes, or come closer together?
 - Did the athlete's ankles stay on the ground, or did they come up on the balls of their feet?
- What to do based on the results:
 - **If your arms came forward as you descended & you couldn't maintain or get into an overhead position:**
 - This would send us down a similar path as if you scored an Average or Below Average on the shoulder mobility assessment. The exercises listed above will help the swimmer build resilient shoulders that can maintain an overhead position.
 - **If their mid back rounded as they descended:**
 - This is a sign they need to work on thoracic extension (when your mid back leans backwards). The best exercises to train this motion will be:
 - Thoracic extension w/ foam roller
 - Thoracic extension w/ overhead raise
 - Thoracic extension w/ pvc press
 - **If their knees came together as they descended:**
 - They can work on the strength of their hips and ankle mobility.
 - To strengthen the hips:
 - Proper technique on squats, split squats & lunges, glute bridge variations, wall sits
 - To improve ankle mobility
 - Foam roll the calves, training with a full range of motion on squats & lunges (go as far down as you can without compromising technique)
 - **If they came up on the balls of their feet as they descended**
 - To improve ankle mobility
 - Foam roll the calves, training with a full range of motion on squats & lunges (go as far down as you can without compromising technique)

- **3. Lying Straight Leg Raise** - This will measure the flexibility of the hamstrings, which goes towards the athlete's overall hip mobility and how well they can move with their lower body. This test is scored on the following criteria:
 - Good - The ankle bone of the raised leg passes the mid-thigh point of the flat leg
 - Average - The ankle bone of the raised leg is somewhere between the knee & mid thigh point of the flat leg
 - Below average - The ankle bone does not reach the knee of the flat leg
- **If you scored a Good:**
 - We'd recommend you be consistent with warming up the lower body musculature before dryland & practice, so you can properly utilize & maintain the lower body flexibility you currently have
- **If you scored an Average or Below average**
 - Aside from doing a proper dynamic warmup before training, here are some lower body mobility/flexibility exercises that will help improve your flexibility:
 - Seated hamstring stretch, pigeon stretch, lying figure 4 stretch, inch worms, downward dog, 90-90 hip switch, hip hinge holds
- **4. Inline Split Squat** - This assessment measures the stability of your lower body. Your ability to maintain a desired body position out of the water with your legs & hips will translate to a better body position in the water coming off the wall & when you're swimming. This test is scored on the following criteria:
 - Good - The athlete maintained their alignment & their stability
 - Average - The athlete's front knee wobbled and/or their torso leaned in a way where they were trying to maintain their balance. The movement was completed, but it looked shaky
 - Below average - The athlete lost their balance and subsequently had to reposition their feet, touch the ground with their hands, etc
- **If the athlete scored an Average or Below average**
 - These exercises will help the athlete improve their balance & stability:
 - One foot balance, peripheral balance, scanning one foot balance, single leg halos, split stance halos, lateral hops, front & back hops, 5 hops to rotational bound, 2 side shuffles to MB chop, pallof hold variations